

Let's Encrypt HTTPS — Integration Status

Revision: 4 **Last modified:** 2026-07-02T16:33:00Z **Status:** In progress — Phases 0-3 + 5 PASS: a REAL hermetic DNS-01 certificate is issued by local Pebble AND auto-renewed/rotated (S1→S2, new serial, zero-downtime swap), both cert-analyzer-verified + re-runnable + guarded. All autonomous LE phases are DONE. **The live-HTTPS half — Phase 4 (LE staging issuance) + Phase 6 (production cutover) — is OPERATOR-BLOCKED, reason "operator-deferred DNS (2026-07-02)"** (the operator deferred live DNS on 2026-07-02; unblock = operator-provided LE account + DNS credentials). The hermetic Phase-3/5 issuance/rotation guards remain GREEN — only the live-cert half is deferred. **Authority:** Inherits constitution/Constitution.md per §11.4.35. §11.4.45 integration-status doc for the Let's Encrypt HTTPS workstream (task #59). **Companion:** design ../LETSENCRYPT_HTTPS.md · plan ../LETSENCRYPT_HTTPS_PLAN.md · summary Status_Summary.md

Operator-blocked items (read first — §11.4.45 O(1) surface)

Item	Why blocked	Unblock condition
Phase 4 — LE staging end-to-end (real DNS-01)	operator-deferred DNS (2026-07-02) — the operator deferred live DNS on 2026-07-02. Also needs a real DNS provider + an API token stored as a Podman secret (§11.4.10 — value never in git/.env); the operator chose <i>defer real domain</i> at Phase 0 and <i>defer live DNS</i> on 2026-07-02.	Operator provides an LE account + DNS credentials (DNS provider name + a scoped API token) and names a test hostname.
Phase 6 — production cutover	operator-deferred DNS (2026-07-02) — depends on Phase 4; real public domain, real DNS-01 for that domain,	Operator authorises the go-live with the real domain + LE account +

Item	Why blocked	Unblock condition
	single operator-gated go-live (design §9).	DNS token + host firewall opening.

Operator-Block-Details (§11.4.21):

- **WHAT:** land the live-HTTPS half — Phase 4 (real LE-staging DNS-01 issuance) then Phase 6 (production cutover). Blocked as OPERATOR-BLOCKED, reason "**operator-deferred DNS (2026-07-02)**".
- **WHY (self-resolution exhausted §11.4.21):** (a) the autonomous half is already DONE — hermetic Phase-3 issuance + Phase-5 renewal/rotation are cert-analyzer-verified, re-runnable, and guarded (GREEN); (b) live issuance requires an LE account + real DNS-provider API credentials that only the operator holds and that MUST NOT be fabricated (§11.4.10 / §11.4.6); (c) the operator explicitly deferred live DNS on 2026-07-02, so this is an operator decision, not an agent gap; (d) no captured fallback / topology substitute exists for a real public-CA cert on a real domain.
- **UNBLOCK CONDITION:** operator provides an LE account + DNS credentials (DNS provider + scoped API token) and names the test/production hostname; then Phase 4 (staging) → Phase 6 (production) can be captured.
- **WHO:** operator (milos85vasic.3rd@gmail.com).

Operator decisions captured (Phase 0 — §11.4.66)

Decision	Choice	Date
TLS client / terminator	Caddy auto-HTTPS (in-process ACME client + terminator; no cron/CI/hook renewal — §11.4.156)	2026-07-01
ACME challenge type	DNS-01 (works without inbound :80/:443 to the host)	2026-07-01
Rollout order	Hermetic (Pebble) + LE-staging first; defer real domain — prod is one later operator-gated cutover	2026-07-01

Phase status (captured-evidence-driven — §11.4.5/§11.4.69)

Phase	Scope	Validation	Evidence
0 — Operator decisions + scaffolding	Capture the 3 design §9 decisions; scaffold the plan	PASS	decisions table above; committed LETSENCRYPT_H + _PLAN.md @ fddd25c
1 — Cert- analyzer (client/ challenge- agnostic)	Pure offline analyzer over a resulting PEM (cert_not_expired / cert_days_remaining / cert_san_matches / cert_chain_roots_in / cert_renewal_due) + golden-good/golden-bad self-validation (§11.4.107(10))	PASS	selftest 37/37; guard GREEN+RED PASS; §1.1 mutation (SAN exact→substring) → guar restore byte-identical mo a06e0f89...; qa-results/ letsencrypt/cert-analyze results/regression/ cert_analyzer_selfvalida
1 — Custom Caddy image w/ DNS-01 provider module	Build a rootless Caddy image bundling the local dns.providers.challtestsrv module (§11.4.161)	PASS	build.sh built localhost/ helix_proxy/caddy- challtestsrv:2.8.4; post- anti-bluff check caddy li modules grep dns.providers.challtests (a build ≠ module linked §11.4.6); qa-results/ letsencrypt/build/
2 — Compose service + Podman secret + cert volume + CoreDNS SOA front	Wire Caddy + CoreDNS (zone-determination fix) + secret NAME/PATH (no value) + persistent cert volume	PASS	compose.hermetic.yml (4 s PEBBLE_VA_ALWAYS_VALID=0, secrets commented for hermetic; caddy loopbac bound per security-audit CoreDNS answers hermet SOA, forwards _acme-cha TXT to challtestsrv
3 — Hermetic local- ACME with Pebble (anti-bluff core)	Caddy issues a REAL cert against local Pebble via DNS-01, fully offline; cert-analyzer asserts the served leaf	PASS	re-runnable phase3_hermetic_issue.sh clean runs each issued a cert (Pebble VA_ALWAYS_VALID=0 g validation); cert-analyzer not_expired + SAN proxy.hermetic.test + ne SAN-reject + chain-to-t run's-Pebble-CA ; qa-res

Phase	Scope	Validation	Evidence
4 — LE staging end-to-end (real DNS-01)	Real LE staging endpoint + real DNS-01	OPERATOR-BLOCKED — reason "operator-deferred DNS (2026-07-02)"	letsencrypt/phase3_issueat 20260701T100119Z/ + ...T100119Z/ operator deferred live D 2026-07-02; unblock = operator-provided LE ac + DNS credentials (§11.4.21)
5 — Renewal + rotation simulation (zero-downtime)	Force the ARI renewal window → Caddy renews via the ACME renewal path → analyzer confirms the new cert; 0 dropped across the swap	PASS	re-runnable phase5_rotat S1→S2 (new serial, later notBefore), caddy log certificate needs renewal on ARI window→renewed successfully, swap ok=6 cert-analyzer PASS (S2→Pebble CA); guard phase5_rotation_guard.sh RED+GREEN proven (w run-tests.sh); qa-results/ letsencrypt/phase5_rotat Trigger mechanism (stor surgery on the cached A restart) is source-justified docs/research/ caddy_2110_ari_refetch_2 — in production Caddy r on its own schedule, no t needed
6 — Production cutover + rollback runbook	Real domain go-live	OPERATOR-BLOCKED — reason "operator-deferred DNS (2026-07-02)"	depends on Phase 4; ope go-live gate (design §9) - operator-provided LE ac + DNS credentials

Honest boundary (§11.4.6)

Phases 0-3 + 5 are DONE with captured evidence. Issuance (Phase 3): the custom Caddy image is built with the DNS-01 module linked (anti-bluff-verified), the hermetic stack is wired (CoreDNS SOA front fixes certmagic's zone-determination), and phase3_hermetic_issue.sh issues a REAL cert via DNS-01 against local Pebble (PEBBLE_VA_ALWAYS_VALID=0) that the cert-analyzer verifies over the served leaf (incl. chain to THIS RUN's Pebble CA). Renewal (Phase 5): phase5_rotation.sh forces the ARI renewal window and Caddy renews via the ACME renewal path

(certificate needs renewal based on ARI window→renewed successfully) to a NEW serial with a later notBefore, the renewal SWAP measured zero-downtime (0 dropped), and the cert-analyzer verifies the new leaf. Both are re-runnable (§11.4.98/ §11.4.107) + guarded (§11.4.135, RED+GREEN). Honest trigger boundary (§11.4.6): the hermetic renewal is forced via storage-surgery on the cached ARI window + a Caddy restart because certmagic v0.21.3 (and even Caddy ≥2.11.0 — source-proven docs/research/caddy_2110_ari_refetch_20260701/) offers no on-demand ARI re-fetch; in PRODUCTION Caddy renews on its own maintenance schedule with no trigger + no restart. NOT exercised (honestly): the real-domain / live-HTTPS paths — Phase 4 (LE staging) + Phase 6 (production cutover) — are OPERATOR-BLOCKED with reason **"operator-deferred DNS (2026-07-02)"** (the operator deferred live DNS on 2026-07-02); they wait on an operator-provided LE account + DNS credentials, never a metadata-only PASS (§11.4 / §11.4.1). The hermetic Phase-3/5 guards stay GREEN — only the live-cert half is deferred.